

DAT THANH NGUYEN

🌐 <https://dattnghuyenx.github.io/aboutme>

✉ datthanhnghuyenx@gmail.com | [in linkedin.com/in/dattnghuyenx](https://www.linkedin.com/in/dattnghuyenx) | ☎ (+49) 1525 581 5369

Graduate research assistant with **5+ years** of experience working on **neural compression, 3D processing, and machine learning**. Proven success bridging academic innovation and real-world application at **Meta Reality Labs** and leading research labs in Europe. Author of **10+** peer-reviewed papers with **320+** citations, open-source implementations with **100+** **GitHub stars**.

EDUCATION

Jul 2025 **Ph.D., Electrical Engineering** UNIVERSITY OF ERLANGEN-NUREMBERG, GERMANY
(expected) Focus: 3D neural compression and rendering | Advisor: Prof. Dr.-Ing. André Kaup

Nov 2020 **M.Sc., Electrical Engineering** POLYTECHNIC INSTITUTE OF PARIS, FRANCE
Focus: Multimedia Networking | GPA: 16.7/20 (Highest Honors)

Mar 2019 **B.Eng., Electrical Engineering** HUST UNIVERSITY, VIETNAM
Scholarships for Excellent Students | GPA: 3.43/4.0 (Top 2%)

EMPLOYMENT AND RESEARCH EXPERIENCE

Jun 2021 **Graduate Research Assistant** UNIVERSITY OF ERLANGEN-NUREMBERG, GERMANY

Jul 2025 Leading the design of ML-based multimodal point cloud coding systems optimized for real-life performance in medical and real-time 3D visualization applications

- **Lossless coding:** Achieved up to $13\times$ compression rate using an efficient hybrid octree-point representation and a generative model that jointly learns point cloud geometry and attribute density, outperforming SOTA approach with 22.6% in bitrate reduction [1, 2]. Improved runtime by $7.5\times$ by later introducing a multiscale context model.

- **Lossy coding:** Achieved up to $480\times$ compression rate with VAE framework adapted for irregular and sparse 3D data. Integrated a hybrid context model capturing local and global patterns, introduced synthetic attribute dataset, resulting in 19.6% bitrate savings and $4\times$ speedup over MPEG G-PCC at equal visual quality [3, 4].

- **Applications:** Collaborated with the capture, visualization, and analysis teams to develop compression pipelines optimized for real-time rendering of huge point clouds, enabling scalable streaming and emotion-aware analytics in human-centered environments [3, 5, 6].

Jan 2023 **Research Scientist Intern** META REALITY LABS, USA

Apr 2023 • Developed and benchmarked mesh simplification algorithms — including edge collapse and vertex clustering — to accelerate rendering and reduce memory overhead in large-scale 3D models.

- Developed a real-time adaptive Level of Detail (LoD) rendering engine for Meta headsets, enabling smooth navigation of complex scenes (28M triangles at 40 FPS) with high visual fidelity.

May 2020 **Master's Intern & Research Engineer** L2S, CENTRALESUPÉLEC, FRANCE

May 2021 Focused on accurate voxel occupancy modeling for point cloud geometry compression.

- Built the first autoregressive voxel occupancy model using 3D masked convolutional networks, reducing bitrate by 28% over existing state-of-the-art methods [7].

- Boosted compression efficiency with a rate-driven partitioning strategy and adaptive context selection, achieving an additional 9% bitrate reduction [8].

- Addressed runtime bottlenecks of previous works with a multiscale coarse-to-fine framework and grouped voxel modeling, enabling parallel, fast encoding/decoding and reducing bitrate by 17.5% on average versus MPEG G-PCC [9].

- Dec 2018 **Deep Learning Engineer** TRI NAM TDI., JSC / FPT SOFTWARE, VIETNAM
- Aug 2019 • Built and deployed real-time face recognition system with high accuracy for smart city applications.
• Integrated fire and smoke detection, traffic violation model into live monitoring platform.
- Dec 2016 **Undergraduate Research Assistant** EMBEDDED SYSTEM LABORATORY, VIETNAM
- Aug 2018 • Designed a novel architecture incorporating adaptive streaming and Software-Defined Networking in video streaming [10].
• Designed and integrated embedded systems with hands-on experience in hardware setup and logic development with FPGAs.

TECHNICAL SKILLS

Programming: C++, Python, Matlab

Frameworks: PyTorch, TensorFlow, Keras, OpenGL, Open3D, PyntCloud, OpenCV

Tools: Git, Linux, Docker, SLURM, LaTeX

AWARDS AND SCHOLARSHIPS

Selected for Heidelberg Laureate Forum 2024 (top 200 globally)
FAU President's Welcome Award (2021)
Labex Digicosme Scholarship (2019)
Erasmus+ Scholarship, TUM (2018)
Scholarships for Excellent Students (2015-17)

ACTIVITIES

Presentations: ICASSP'21, '23; ICMEW'21; ICIP'22, '24; Asilomar'24; INISCOM'19
Reviewer: IEEE T-CSVT, TIP, VCIP, ICASSP, Image Communication
Mentor: led Python-based Image Processing course, advisor to 3 Master's and 2 Bachelor's theses

Selected Publications

- [1] D. T. Nguyen and A. Kaup, "Lossless point cloud geometry and attribute compression using a learned conditional probability model," *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
- [2] D. T. Nguyen and A. Kaup, "Learning-based lossless point cloud geometry coding using sparse tensors," in *Proceedings of the IEEE International Conference on Image Processing*, 2022.
- [3] D. T. Nguyen, M. Sadeghi, D. Zieger, M. Berking, B. M. Eskofier, M. Stamminger, and A. Kaup, "Region-of-interest based human body point cloud attribute compression," *Submitted to IEEE Transactions on Image Processing*, 2025.
- [4] D. T. Nguyen, D. Zieger, M. Stamminger, and A. Kaup, "End-to-end learned lossy dynamic point cloud attribute compression," in *Proceedings of the IEEE International Conference on Image Processing*, 2024.
- [5] D. T. Nguyen, D. Zieger, M. Gambietz, A. D. Koelewijn, M. Stamminger, and A. Kaup, "Multiresolution point cloud compression for real-time visualization and streaming of large 3d datasets," in *Proceedings of the 2024 58th Asilomar Conference on Signals, Systems, and Computers*.
- [6] D. T. Nguyen, K. G. Nambiar, and A. Kaup, "Deep probabilistic model for lossless scalable point cloud attribute compression," in *Proceedings of the 2023 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2023.
- [7] D. T. Nguyen, M. Quach, G. Valenzise, and P. Duhamel, "Learning-based lossless compression of 3d point cloud geometry," in *ICASSP 2021 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2021.
- [8] D. T. Nguyen, M. Quach, G. Valenzise, and P. Duhamel, "Lossless coding of point cloud geometry using a deep generative model," *IEEE Transactions on Circuits and Systems for Video Technology*, 2021.
- [9] D. T. Nguyen, M. Quach, G. Valenzise, and P. Duhamel, "Multiscale deep context modeling for lossless point cloud geometry compression," in *2021 IEEE International Conference on Multimedia Expo Workshops (ICMEW)*, 2021.
- [10] P. H. Thinh, . Dat, Nguyen Thanh, and T. T. Huong, "An efficient qoe-aware http adaptive streaming over software defined networking," *Mobile Networks and Applications*, 2020.